

Uday Kamal

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SUMMARY	6 years of experience in developing and applying Deep Learning (CNNs, Transformers) methods to solve diverse computer vision and robotics research problems.	
EDUCATION	Georgia Institute of Technology Atlanta, GA PhD in Electrical and Computer Engineering (ML) Jan. 2021 – Apr. 2025 Advisor: Saibal Mukhopadhyay Georgia Institute of Technology Atlanta, GA MS in Electrical and Computer Engineering Jan. 2021 – Dec 2022 Bangladesh University of Engineering & Technology Bangladesh BS in Electrical and Electronics Engineering Feb. 2015 – Apr 2019	
RELEVANT EXPERIENCE	GAM: Generalized Action Model for Robotic Manipulation • Developed a production-grade Vision-Language-Action (VLA) foundation model that unifies diverse robotic manipulation actions—including suction, pinch grasps, caging, and placements—into a single language-prompted policy and critic architecture. • Validated system reliability across a fleet of physical robotic work-cells over 10 million industrial pick-and-place cycles under highly cluttered scenes. • Achieved greater than 95% success rates for autonomous grasp and placement action generation, demonstrating highly robust multi-task cross-embodiment generalization.	
EXPERIENCE	Memory-Augmented Transformer for Event-based Object Recognition • Developed a novel spatiotemporal representation learning framework that combines external memory mechanism with Attention for streaming event-based perception. • Designed a learnable memory read-and-write mechanism with sliding window attention that enables highly efficient processing of event-based sequences. • Achieved 30000x computation gain compared to the state-of-the-art methods with no loss in accuracy on event-based object recognition datasets. • Demonstrated persistent memory on long horizon sequential classification and robustness across input transformation. Hierarchical Learning with Adaptive Update for Event-based Dense Perception • Developed a hierarchical, multi-stage memory-enhanced Transformer encoder architecture with an adaptive update rate for efficient event-based dense perception. • Proposed a global context-based association between two contiguous memory levels to efficiently process recurrent updates. • Achieved significant (50%) reduction in latency while maintaining competitive performance on different event-based dense perception tasks (e.g. object detection, semantic segmentation and depth estimation).	

Reinforcement Learning with Event-Based Perception

- Developed a novel perception module for long-horizon sequential decision-making tasks with adaptive action-taking frequencies.
- Designed an integrated, event-by-event reinforcement learning framework for end-to-end policy learning conditioned on optimal event-based representations for autonomous navigation and obstacle avoidance.
- Developed a recurrent action gating mechanism to execute actions at an adaptive frequency, reducing compute cost and improving sample efficiency.

SELECT PUBLICATIONS

[7] U. **Kamal**, J. Jiang, C. Mitash, W. Wang, I. Obi-Okoye, C. Wang, A. Jiwani, H. Mao, H. Nguyen, M. Breyer, W. Xia, N. Mishra, K. Bekris, K. Katyal GAM: Generalized Action Model for Robotic Manipulation. *ICRA 2026 Workshop*.

[6] M. S. Mohammad, U. **Kamal**, S. Mukhopadhyay. Adaptive-Cloud: Dynamic Computation Control for 3D Object Detection From LIDAR Point Clouds Using Matryoshka Representation Learning. *IEEE RA-L 2025*.

[5] U. **Kamal**, S. Mukhopadhyay. Memory-Augmented Representation for Efficient Event-based Visuomotor Policy Learning with Adaptive Perception and Control. *WACV 2026*.

[4] U. **Kamal**, S. Mukhopadhyay. Efficient Learning of Event-based Dense Representation using Hierarchical Memories with Adaptive Update. *ECCV 2024*.

[3] M. Lee, U. **Kamal**, S. Mukhopadhyay. Learning Collective Dynamics of Multi-Agent Systems using Event-based Vision. *L4DC 2025*.

[2] U. **Kamal**, S. Dash, S. Mukhopadhyay. Associative Memory Augmented Asynchronous Spatiotemporal Representation Learning for Event-based Perception. *ICLR 2023 (Notable-25%)*.

[1] U. **Kamal**, T. I. Tonmoy, S. Das, Md. K. Hasan. Automatic Traffic Sign Detection and Recognition Using SegU-Net and a Modified Tversky Loss Function with L1-Constraint. *IEEE T-ITS 2017*.

WORK EXPERIENCE

Applied Scientist II, Amazon Robotics, Boston, MA

May 2025 - Current

Applied science full-time position with the Robotic Manipulation Group under Dr. Kapil Katyal. Developing foundation VLA models using score-based generative learning followed by RL-based post-training to solve some of the most challenging industry-scale robotics problems including warehouse items pick and place, motion planning and continuous learning.

EXPERIENCE

Applied Scientist Intern, Amazon Robotics, Boston, MA

Aug - Dec 2023

Applied science internship with the Robotic Manipulation Group under Dr. Chaitanya Mitash and Dr. Jeroen Van Baar. Led research project on novel 3D-scene graph augmented semantic scene understanding algorithm to predict pick outcome for target object under heavily cluttered environments with warehouse scale robots. Also collaborated with fellow intern on diffusion-based visuo-motor policy learning integrated with 3D perception for scooping from cluttered bins with real robots.

Research Engineer, Brain Station 23 Limited, Bangladesh

Aug - Dec 2020

Research engineer with the intelligent biomedical imaging research group under Dr. Taufiq Hasan. Led research project on novel anatomy-aware spatial attention and leveraged generative data-augmentation (using Cycle-GAN architecture) in a semi-supervised learning setting to tackle imperfect and scarce data annotation.

Research Associate, Neural Semiconductor, Bangladesh

Jun 2019 - May 2020

Research associate with the advanced hardware acceleration and digital design group under Dr. A.B.M. Harun-ur Rashid. Led the project on developing an end-to-end HLS-based FPGA hardware accelerator compiler that supports several building blocks including pooling, convolution and linear layers for accelerated inference of quantized deep CNN architectures (VGG16, and TinyYOLO) and prototyped them on real FPGA board (Xilinx Ultra96)

AWARDS

& GRANTS

Silver medal in Kaggle APTOS Blindness Detection Challenge	2019
2nd place in IEEE SPS Video and Image Processing Cup	2018
Travel grant for International Conference in Image Processing (ICIP)	2018
1st place in IEEE SPS Video and Image Processing Cup	2017
Travel grant for International Conference in Image Processing (ICIP)	2017
Undergraduate Deans Award	2016
Government Scholarship for academic excellence in Higher Secondary College	2011